1. **Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.**

Step 1: Connect to Wi-Fi
Connect your Windows laptop and your friend's phone to the same Wi-Fi network.

Step 2: Find the laptop's IP address
Open Command Prompt on your laptop and type:

```
ipconfig
```

Note the IPv4 address under the Wi-Fi adapter.

Step 3: Find the phone's IP address
On the phone, open Settings → Wi-Fi → Connected network details and note its IP address.

Step 4: Ping the phone from the laptop
In Command Prompt, type:

```
ping 192.168.1.25
```

Replace 192.168.1.25 with your friend's actual phone IP address.

Step 5: Check the result

- Reply from… means the phone is responding to ping requests.
- Request timed out means the phone did not respond. Check Wi-Fi settings and whether the phone or router blocks ping requests.

Step 6: Record the result
Write down the ping result and whether communication was successful.

2. Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

Step 1: Open Command Prompt on your Windows laptop.

Step 2: Type the following command and press Enter.

```
tracert www.youtube.com
```

Step 3: Wait for the command to complete. The output displays the network hops between your laptop and the destination.

Step 4: Count the numbered hop lines in the output to find how many hops responded. Asterisks (`*`) indicate that a hop did not respond within the timeout.

Observation:

- Destination: www.youtube.com
- Command used: `tracert www.youtube.com`
- Number of hops: _____ (fill in after running the command)

Result: The traceroute command was used to examine the network path from the laptop to YouTube. The actual number of hops depends on the network connection and routing at the time of testing.

3. Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

Step 1: Connect both devices to Wi-Fi.
Connect your Windows laptop and mobile phone to the same Wi-Fi network.

Step 2: Find the mobile phone's IP address.
Open the phone's Wi-Fi settings, select the connected network, and note its IP address. For example, 192.168.1.25.

Step 3: Open Command Prompt.
On your laptop, press Windows + R, type cmd, and press Enter.

Step 4: Test the connection.
Type the following command, replacing the example IP address with your phone's actual IP address:

ping 192.168.1.25

If the phone responds, you may see Reply from 192.168.1.25.

Step 5: Disconnect the phone.
Turn off Wi-Fi on the mobile phone so it is no longer connected to the same network.

Step 6: Ping the phone again.
Run the same command in Command Prompt:

ping 192.168.1.25

Step 7: Observe the error message.
A common result is:

Request timed out.

This means the laptop did not receive a response from the phone within the allowed time. Depending on the network, you might instead see Destination host unreachable.

Observation

Test	Expected result
Phone connected to Wi-Fi	Reply may be received
Phone disconnected from Wi-Fi	Request timed out or host unreachable
Reason	The phone is no longer reachable through that Wi-Fi connection

Result

After disconnecting the phone from Wi-Fi, the laptop could no longer receive ping replies from it. The test demonstrates how the ping command helps check network connectivity between devices.

4. Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

Step 1: Connect your laptop to your home Wi-Fi network.

Step 2: Open Command Prompt by pressing Windows + R, typing cmd, and pressing Enter.

Step 3: Type the following command:

arp -a

Step 4: Note the IP addresses and corresponding physical (MAC) addresses displayed in the output.

Step 5: Identify each device by checking its IP address in your router's connected-device list or by comparing MAC addresses.

Observation Table

Sr. No.	IP Address (Example)	Device / Person
1	192.168.1.1	Wi-Fi router
2	192.168.1.2	My laptop
3	192.168.1.3	Father's mobile phone
4	192.168.1.4	Mother's mobile phone
5	192.168.1.5	Smart TV

5. Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.Include screenshots or command outputs if possible.

1. Objective

To set up a local Wi-Fi network, identify connected devices, and test network connectivity using the ping, tracert, and arp -a commands.

2. Procedure

Step 1: Connect Devices to Wi-Fi
I connected my Windows laptop and mobile phone to the same home Wi-Fi network. I checked that both devices were connected successfully.

Step 2: Find IP Addresses
I opened Command Prompt on my laptop and entered ipconfig to find the laptop's IPv4 address, subnet mask, and default gateway. I checked the phone's Wi-Fi settings to find its IP address.

Step 3: Test Connectivity Using Ping
I entered ping [phone IP address] in Command Prompt to check communication between the laptop and phone. When a reply was received, it indicated that the phone was responding to ping requests.

Step 4: Test the Network Route
I executed tracert www.youtube.com to examine the route taken by network packets to reach YouTube. The output displayed numbered hops, response times, and some hops that did not respond.

Step 5: Test After Disconnecting a Device
I disconnected the phone from Wi-Fi and repeated the ping command. The expected result was Request timed out because the phone was no longer reachable through that Wi-Fi connection.

Step 6: Identify Network Devices
I ran arp -a to view IP and MAC address entries stored in the laptop's ARP cache. I used the information to help identify devices on the local network.

3. Results and Observations

Test	Observation
Wi-Fi connection	Laptop and phone connected to the same network.
Ping test	Replies indicated successful communication when the phone responded.
Traceroute test	Displayed the network hops towards YouTube.
Disconnected device	Ping could time out when the phone was disconnected.
ARP command	Displayed IP and MAC address entries available in the ARP cache.

4. Issues Observed

Some devices may not respond to ping because of firewall settings. In traceroute, some hops may show asterisks (*) if routers do not respond within the timeout. The arp -a command may not list every device connected to the Wi-Fi network.

5. Conclusion

The practical helped me understand how to configure and inspect a Wi-Fi network and test connectivity using Windows networking commands. I learned how ping checks device responsiveness, tracert displays the network route, and arp -a shows locally cached IP-to-MAC address mappings.